

Co⁺ ionisation limit 63564.60 cm⁻¹ = 7.88101 eV (157.320 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle V | \text{er} | J' \rangle$ in a_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$
 Level caption: NIST energy and measured lifetime (where one exists).



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

⁵⁹Co hyperfine structure ($I = 7/2$)

level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)							
$^4F_{5/2}$	450.2835(9)	139.22(3)	F=1: -8604.15	F=2: -7723.47	F=3: -6398.31	F=4: -4623.69	F=5: -2392.99	F=6: +302.08	F=7: +3471.47	F=8: +7126.77
$^4F_{3/2}$	490.5678(21)	94.50(3)	F=0: -7675.82	F=1: -7194.90	F=2: -6231.12	F=3: -4780.62	F=4: -2837.64	F=5: -394.44	F=6: +2558.61	F=7: +6033.08
$^4F_{1/2}$	613.348(3)	67.54(6)	F=1: -6863.99	F=2: -5556.59	F=3: -3838.25	F=4: -1400.30	F=5: +1668.86	F=6: +5383.68		
$^4F_{3/2}^o$	1042.9828(9)	67.618(18)	F=2: -7003.91	F=3: -3923.26	F=4: +229.35	F=5: +5492.56				
$^4F_{5/2}^o$	828.800(3)	-118.7(3)	F=1: -16008.82	F=2: -14334.26	F=3: -11825.95	F=4: -8488.14	F=5: -4326.47	F=6: +651.99	F=7: +6438.75	F=8: +13023.92
$^4F_{3/2}^o$	668.918(3)	-79.23(21)	F=0: -10577.91	F=1: -9900.90	F=2: -8548.51	F=3: -6523.97	F=4: -3832.13	F=5: -479.45	F=6: +3525.97	F=7: +8174.44
$^4F_{1/2}^o$	810.0(4)	-60.00	F=1: -15620.77	F=2: -13992.12	F=3: -11550.93	F=4: -8299.34	F=5: -4240.21	F=6: +622.89	F=7: +6285.67	F=8: +12743.13
$^4G_{5/2}$	655.4(24)	30.00	F=0: -10305.69	F=1: -9653.40	F=2: -8348.21	F=3: -6388.90	F=4: -3773.62	F=5: -499.93	F=6: +3435.23	F=7: +8035.54
$^4G_{3/2}^o$	772.0(21)	210.00	F=2: -17475.45	F=3: -15193.91	F=4: -12144.21	F=5: -8319.81	F=6: -3712.54	F=7: +1687.43	F=8: +7891.56	F=9: +14912.92
$^4G_{1/2}^o$	559.1(21)	30.00	F=1: -6273.92	F=2: -5164.27	F=3: -3496.58	F=4: -1267.00	F=5: +1529.62	F=6: +4899.71		
$^4G_{5/2}^o$	517.1(27)	180.00	F=1: -9872.44	F=2: -8863.88	F=3: -7345.67	F=4: -5311.40	F=5: -2752.49	F=6: +341.78	F=7: +3984.26	F=8: +8189.95
$^4D_{5/2}^o$	751.0(27)	-	F=0: -11827.93	F=1: -11076.95	F=2: -9575.00	F=3: -7322.06	F=4: -4318.14	F=5: -563.24	F=6: +3942.64	F=7: +9199.50
$^4G_{3/2}^o$	448.2(15)	150.00	F=0: -6978.64	F=1: -6545.75	F=2: -5676.92	F=3: -4366.03	F=4: -2603.88	F=5: -378.23	F=6: +2326.21	F=7: +5527.83
$^4D_{3/2}^o$	696.1(27)	30.00	F=1: -7815.28	F=2: -6431.61	F=3: -4352.89	F=4: -1575.27	F=5: +1906.40	F=6: +6098.55		
$^4D_{1/2}^o$	398.4(18)	150.00	F=1: -4401.87	F=2: -3647.89	F=3: -2500.84	F=4: -941.45	F=5: +1056.01	F=6: +3523.68		
$^4D_{5/2}^o$	705(6)	30.00	F=2: -4739.37	F=3: -2647.27	F=4: +162.20	F=5: +3706.18				
$^4G_{5/2}^o$	495(3)	-	F=1: -9522.20	F=2: -8532.89	F=3: -7048.90	F=4: -5070.27	F=5: -2596.97	F=6: +371.00	F=7: +3833.61	F=8: +7790.90
$^4F_{3/2}^o$	448.8(18)	180.00	F=0: -6972.01	F=1: -6541.59	F=2: -5617.07	F=3: -4371.11	F=4: -2612.68	F=5: -387.10	F=6: +2324.01	F=7: +5542.68
$^4D_{5/2}^o$	478.2(4)	30.00	F=0: -7450.82	F=1: -6987.96	F=2: -6059.17	F=3: -4658.33	F=4: -2776.26	F=5: -400.72	F=6: +2483.61	F=7: +5895.08
$^4G_{1/2}^o$	300(6)	-	F=0: -6820.22	F=1: -5920.85	F=2: -4721.69	F=3: -3222.74	F=4: -1424.00	F=5: -674.53	F=6: +3207.85	F=7: +5770.96
$^4G_{3/2}^o$	334.3(27)	-	F=1: -6434.70	F=2: -5766.16	F=3: -4763.35	F=4: -3426.27	F=5: -1754.92	F=6: +250.70	F=7: +2590.59	F=8: +5264.75
$^4D_{5/2}^o$	465(6)	-	F=1: -5227.65	F=2: -4298.29	F=3: -2904.25	F=4: -1045.53	F=5: +1277.87	F=6: +4065.95		
$^4F_{5/2}^o$	297(3)	240.00	F=1: -5603.21	F=2: -5043.91	F=3: -4197.83	F=4: -3056.38	F=5: -1608.15	F=6: +161.16	F=7: +2268.69	F=8: +4734.44
$^4F_{3/2}^o$	508.1(2)	-	F=0: -8003.36	F=1: -7495.21	F=2: -6478.91	F=3: -4954.46	F=4: -2921.86	F=5: -381.11	F=6: +2667.79	F=7: +6224.8

Measured A, B constants; sources in the data file.